we're using the boldface **P** and **X** because we plan to focus on the quantum mechanical Hamiltonian.

Let's start by doing a manipulation that is correct for classical physics but will require some modification for quantum mechanics. In the parentheses above, we have a sum of squares. Using the formula

$$a^2 + b^2 = (a + ib)(a - ib),$$

it seems that we can rewrite the Hamiltonian as

$$\mathbf{H} \text{ “}=\text{”} \frac{1}{2}(\mathbf{P} + i\omega\mathbf{X})(\mathbf{P} - i\omega\mathbf{X}), \tag{10.19}$$

and that's almost correct. Why *almost*? Because quantum mechanically, **P** and **X** do not commute, and we need to be careful about the order of operations. Let's expand our factored expression and see how it might differ from the original Hamiltonian in Eq. 10.18. Keeping careful track of the order of factors, we can expand the expression as follows:

$$\begin{aligned}
\frac{1}{2}(\mathbf{P} + i\omega\mathbf{X})(\mathbf{P} - i\omega\mathbf{X}) &= \frac{1}{2}(\mathbf{P}^2 + i\omega\mathbf{X}\mathbf{P} - i\omega\mathbf{P}\mathbf{X} - i^2\omega^2\mathbf{X}^2) \\
&= \frac{1}{2}(\mathbf{P}^2 + i\omega(\mathbf{X}\mathbf{P} - \mathbf{P}\mathbf{X}) - i^2\omega^2\mathbf{X}^2) \\
&= \frac{1}{2}(\mathbf{P}^2 + i\omega(\mathbf{X}\mathbf{P} - \mathbf{P}\mathbf{X}) + \omega^2\mathbf{X}^2) \\
&= \frac{1}{2}(\mathbf{P}^2 + \omega^2\mathbf{X}^2) + \frac{1}{2}i\omega(\mathbf{X}\mathbf{P} - \mathbf{P}\mathbf{X}).
\end{aligned}$$

Look at the right-hand set of parentheses in the final line. We have seen that expression before—it's the commutator of **X** and **P**. In fact, we already know its value: